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Extension & Execution

***Owns:** Manifest V3 skeleton, content scripts, background service worker, action executor, client action validator.*

Role 1 of 6 · EXTENSION

HW Hardware access

None required — this role's entire workload runs on any laptop.

No booking needed for your own work. You may briefly borrow Laptop A/B to sanity-check execution under Perception's live model load, but that's their slot, not yours.

P0 Phase 0 — now → 20 Sep 2026 (lock the idea)

- Set up the **extension/** folder skeleton — manifest.json, content script, background service worker stub.
- Get a “hello world” content script injecting into the mock site the moment Integration has it up.
- Agree the message-passing contract (content script ↔ background ↔ popup/dashboard) with Perception and Privacy Guard — they both hook into it.

P1 Phase 1 — build the MVP (20 Sep → shortlist)

- Build the Manifest V3 skeleton: permissions, messaging, basic popup/inspector UI shell.
- Implement the **client action validator** — target element exists, action type is allowed, tab/session matches, confidence threshold met.
- Implement safe action execution mapping structured actions (click / type / scroll / navigate / keypress / wait) to real browser APIs.
- Build the **type_secret** pattern — the server requests a semantic operation, the real secret value is resolved locally and never crosses the network.
- Handle the MV3 service-worker lifecycle correctly: checkpoint state to storage rather than in-memory globals — it terminates after ~30s idle with no warning.

P2 Phase 2 — rehearse & harden (shortlist → Dec finale)

- Stress-test the executor against a page that changes mid-session (the “the button moved” case) — this is the first thing a judge will try.
- Rehearse the recovery path with PVM/Verification: failed/ambiguous action → capture new state → re-request reasoning → retry safely.

DEP Dependencies

YOU DEPEND ON	Server AI (action schema/response format), PVM & Verification (recovery-loop triggers).
DEPENDS ON YOU	Everyone. You're the runtime host every other module plugs into — if your scaffold slips, all five other roles stall.

Full team plan, hardware-booking rules, and the master phase timeline: see the companion *PS26171 Team Workflow & Hardware Plan* document.